

Real-Time Cognitive Emotion Mapping for Human- Avatar Interaction in VR

Seif Ehab El Sheikh
Media Engineering Technology
German University
Cairo, Egypt
selsheikh04@gmail.com

Hossam Sanad
Media Engineering Technology
German University
Cairo, Egypt
hossamsanadd@gmail.com

Nada Hamed Sharaf
Media Engineering Technology
German International University
Cairo, Egypt
nada.hamed@giu-uni.de

Mohammed A.-M. Salem
Media Engineering Technology
German University
Cairo, Egypt
mohammed.salem@guc.edu.eg

Abstract—With the shift toward online platforms for education, coaching, and therapy especially accelerated by the COVID19 pandemic—the need for emotionally expressive digital tools has grown significantly. Traditional virtual education solutions often lack the emotional interactivity required for effective communication, particularly for children with communication disorders such as autism. Furthermore, most avatar-based systems are built around English-speaking contexts, leaving a cultural and linguistic gap for widely spoken dialects such as Egyptian Arabic. This paper presents the development and evaluation of an Egyptian emotional avatar designed to express six primary emotions: Anger, Sadness, Fear, Surprise, Happiness, and Neutrality at varying intensity levels. The avatar was adapted from a Unity Asset Store model (Awaii Studio), enhanced by adding ten custom blendshapes using Blender to support additional facial expressions. Emotion targets were modeled based on average facial expressions of an Egyptian actor, derived from a curated dataset. Synchronization of lip movement was achieved using an Egyptian Arabic Text-to-Speech model developed by Omar Samir, alongside a blinking system to enhance realism. A user study was conducted with 40 participants (22 female, 18 male), comparing avatar videos with and without emotional expressions. Participants were divided to mitigate order bias and assessed the avatar based on naturalness, understanding, and reasonableness. Filler and attention-check questions were included to ensure reliability of responses. Results showed that 95% of participants preferred the emotionally expressive avatar, and found its expressions believable and contextually appropriate. This research contributes a culturally-tailored, emotionally rich avatar system for Arabic-speaking learners. Future work will explore dynamic emotional transitions, integration of body language, and enhancements to the TTS system for improved Egyptian Arabic fluency and realism.
Index Terms—emotion recognition, avatar, Egyptian Arabic, virtual education, blendshapes, facial expression, VR storytelling

I. INTRODUCTION

The integration of emotionally expressive avatars into digital platforms has gained significant attention in recent years, particularly in the fields of education, healthcare, and assistive technologies. As virtual environments increasingly replace in-person interaction, there is a growing need for avatars capable of conveying nuanced emotions—especially for users with communication difficulties such as autism or ADHD. Emotional expression enhances engagement, comprehension, and user connection, positioning avatars as

essential tools for inclusive and accessible communication. Existing research has explored a variety of modalities for avatar-based emotion rendering, including facial animation, speech synthesis, and biosignal processing. Multimodal systems such as those by Johnson et al. [1] and Wang et al. [8] have shown that expressive avatars can support socialemotional learning and improve human-computer interaction. Techniques like blendshape interpolation and the Facial Action Coding System (FACS) [4] have proven effective for generating nuanced facial expressions, while newer approaches like ECAvatar [9] and EVOKE [6] incorporate deep learning and EEG signals to map internal states to avatar behavior. Educational use cases, particularly involving children with autism, have also been explored [5], [7], showing the impact of emotional realism on learning motivation. More recently, Burleigh et al. [17] evaluated the effectiveness of avatars using blendshape-based emotional expressions to support emotion recognition in children with Autism Spectrum Disorder (ASD), highlighting the role of expression realism in avatar-assisted communication. Despite these advances, most emotionally intelligent avatars are designed for English-speaking contexts and lack adaptation to other languages or cultural frameworks. In particular, there is a clear research gap in the development of expressive avatars tailored for Egyptian Arabic—a dialect spoken by over 100 million people. Existing systems often use generic expressions and rely on English-based speech synthesis, limiting their realism and relatability for Arabic-speaking users. Moreover, synchronization between emotional speech and facial animation is frequently insufficient, resulting in avatars that feel mechanical or disconnected. Advanced techniques such as deepfake-based face reenactment and neural rendering have emerged, but were excluded in this work due to their dependence on photorealistic avatars and limited suitability for stylized or real-time 3D applications. Instead, this project focuses on blendshape-based modeling—widely used in game and animation pipelines—for real-time facial animation, enhanced by speech-driven jaw motion and automated blinking. This paper presents the design, development, and evaluation of a culturally grounded, emotionally expressive 3D avatar capable of storytelling in Egyptian Arabic. The avatar supports six emotions—happy, sad, angry, fear, surprise, and neutral—each rendered at three intensity levels. These expressions

were generated using blendshape interpolation guided by facial data extracted from Egyptian actor footage. An

Egyptian Arabic text-to-speech (TTS) model with emotional modulation was used to synthesize expressive speech, and all expressions were synchronized in real time within a Unity-based system. A user study involving 40 participants was conducted to compare the emotionally expressive avatar to a neutral version using identical scripts. Results demonstrated strong participant preference for the expressive avatar in terms of clarity, realism, and engagement, particularly in educational storytelling contexts.

The main contributions of this work are:

- The creation of an emotionally expressive 3D avatar capable of speaking Egyptian Arabic and expressing six core emotions at three intensity levels.
- Integration of blendshape-based facial animation, emotional TTS, and real-time jaw and blink control within a Unity application.
- Development of an annotated video dataset from Egyptian actors to guide blendshape tuning and support future emotion recognition.
- Design and evaluation of a complete storytelling pipeline, validated through a user study comparing expressive and non-expressive avatar performance.

Paper Organization: Section II describes the overall system architecture and methodology. Section III presents the implementation details of avatar design, dataset creation, and audio-visual synchronization. Section IV discusses the user study and evaluation. Section V summarizes results and their implications. Section VI concludes the paper and outlines future research directions.

II. SYSTEM DESIGN AND METHODOLOGY

The system accepts an emotion-labeled audio input and outputs a 3D animated avatar synchronized to that data.

III. METHODOLOGY

The development of the emotionally expressive Egyptian avatar followed a multi-stage pipeline that included story creation and annotation, emotional speech generation, avatar animation and synchronization, culturally grounded dataset construction, and preliminary implementation of emotion recognition models.

A. Story Creation and Audio Generation

To enable emotionally dynamic storytelling, three short Egyptian Arabic stories were written around the debatable topic “Is online gaming good?”. The topic was chosen to naturally elicit a wide range of emotional tones. Each sentence in the stories was manually annotated with one of six core emotions—happy, sad, angry, fear, surprise, or neutral—and categorized into three intensity levels: **weak**, **moderate**, and **strong**. Speech synthesis was performed using an Egyptian Arabic Text-to-Speech (EGTTS) model based on the XTTS architecture, which allows emotional voice cloning from a short reference sample. The model accepted sentence-level text input along with a reference voice and an emotion-controlling temperature parameter. By adjusting this temperature, the speech output reflected varying emotional intensities. The model was accessed locally via a Python application using Gradio, which provided a simple interface for generating and reviewing the audio outputs. These audio clips were then exported in WAV format for use in avatar animation

B. Avatar Integration and Real-Time Emotion Playback

Each sentence-level audio clip and its corresponding emotion label were passed into Unity, where a modular animation system was built. The emotion labels

(e.g., happy_2) were mapped to pre-defined blendshape configurations that control the avatar’s facial expressions. These blendshapes were created and adjusted in Blender and stored as numerical weights in a CSV file. Inside Unity, two custom C# scripts—AvatarStoryPlayer.cs and EmotionReceiver.cs—were developed. The first script handled sequential playback of audio clips and passed the associated emotion label to the second script. EmotionReceiver.cs applied the corresponding blendshape weights in real time to create synchronized emotional expressions. To enhance realism, the avatar’s jaw movement was dynamically animated using RMS (Root Mean Square) analysis of the audio waveform. This made the avatar appear to speak in coordination with the audio. Additionally, a blinking system was implemented with a fixed interval to avoid visual stiffness during narration. The entire system was built around Unity GameObjects and Audio Sources, where each “storyline” object combined a clip of speech with its matching emotional animation.

C. Final Avatar System

The final avatar is a fully animated 3D character capable of narrating Egyptian Arabic stories with synchronized facial expressions and emotionally appropriate speech. A Skinned Mesh Renderer was used to enable smooth facial deformation through blendshapes, allowing transitions between expressions via linear interpolation. These transitions enabled smooth and natural changes between emotions over time. The combination of emotional facial expressions, speech-synchronized jaw motion, and automatic blinking resulted in a more believable and immersive avatar. Although the current version relies on pre-annotated stories and manually mapped blendshape values, the system architecture was designed to allow future upgrades for real-time emotional response based on detected input.

D. Dataset Preparation

To ensure culturally relevant emotion representation, a custom dataset was built using video clips from two well-known Egyptian actors. The primary source was Ahmed Helmy, chosen for his highly expressive acting and strong cultural resonance. Several of his films and series were reviewed, and over 200 emotionally diverse clips were extracted using LosslessCut, a lightweight video editing tool. This process preserved video quality while enabling precise scene segmentation. Each clip was reviewed and labeled with one of the six emotions and assigned an intensity level. A secondary dataset was collected from Ahmed Ezz, known for more serious roles. However, due to limited emotional variety and lower quality, this dataset was excluded from the current implementation and reserved for future expansion.

E. Preprocessing and Sample Selection

All labeled clips were processed to extract structured features. OpenCV was used to load each video and convert it into frames. MediaPipe FaceMesh detected 468 facial landmarks per frame, allowing for automatic cropping of the

face region. MediaPipe Pose was also applied to extract full-body keypoints. These data were saved along with emotion and intensity labels for each frame. This structured data was used in two ways. First, facial crops were used to guide manual mapping of emotion expressions to blendshape values in Unity. Second, pose keypoints were archived for potential use in training a body language-based emotion recognition model. A curated subset of 15 samples per emotion was selected, totaling 90 clips. Each was mapped manually to a blendshape configuration, then categorized by intensity. The average of these blendshape vectors was calculated per emotion-intensity pair, forming the reference expressions used for avatar animation.

F. Emotion Recognition Models

Three unimodal emotion recognition models were developed as part of a future plan for dynamic avatar control. These models focus on facial expression, speech audio, and body posture. The facial model was trained using a ResNet-18 architecture on the extracted face crops. Images were resized, normalized, and divided using an 80/20 train-test split. The model was trained for 15 epochs using the Adam optimizer and achieved a test accuracy of 58%. The speech emotion model used mel spectrograms generated from Arabic audio clips. These were passed into another ResNet-18 network fine-tuned for emotion classification. Despite a larger dataset, the model achieved 55% accuracy. The body language model used 2D keypoints from MediaPipe Pose, flattened and fed into a fully connected neural network. Due to subtle movement differences and limited variety, this model achieved only 30% accuracy. A fusion system was designed to combine the outputs of the three models using a weighted average. However, given the limited performance of the individual models, the fusion system was not implemented in the final application. These models are retained for future enhancement when higher accuracy datasets and tuning strategies are available.

G. Text Emotion Annotation

To ensure alignment between storytelling content and emotional expression, each story was developed with emotional variability in mind. Topics were selected based on their ability to naturally evoke multiple emotions—such as online gaming, which can involve moments of joy, frustration, surprise, or disappointment. After writing the full story script in Egyptian Arabic, each sentence was manually annotated with an emotion label and intensity level. This annotation ensured that every line of the story could trigger the correct facial expression and vocal tone in the avatar, enabling sentence-level synchronization between narrative and emotion.

IV. AVATAR DESIGN IN UNITY

The avatar used in this project was adapted to support a wide range of facial expressions through blendshape manipulation. This section outlines the process of preparing and extending the avatar to visualize emotional states accurately.

A. Avatar Preparation and Blendshape Extension

The base 3D avatar was sourced from the Unity Asset Store and originally included seven blendshapes, covering

basic facial actions such as jaw movement and eye blinking. However, these were insufficient for conveying nuanced emotional expressions. In 3D animation, blendshapes are used to deform a character's mesh to simulate facial movements, where each blendshape represents a specific expression such as smiling, frowning, or raising eyebrows. The intensity of each expression is controlled by blendshape weights ranging from 0 to 100, allowing for smooth transitions and combinations of multiple expressions. To enhance expressiveness, the avatar model was imported into Blender for modification. Ten additional blendshapes were manually sculpted using Blender's editing tools such as Smooth, Elastic Snake Hook, and Grab. These new expressions included subtle emotional cues like smirks, nose wrinkles, and varied smile intensities. This increased the total number of blendshapes from seven to seventeen, significantly improving the avatar's ability to represent complex and culturally accurate facial emotions. These custom expressions were later controlled in Unity to synchronize with the story's emotional annotations and speech.

B. Manual Emotion Matching and Blendshape Averaging

To establish a reliable reference for avatar expressions, 15 samples were selected for each of the six core emotions—happy, sad, angry, fear, surprise, and neutral—from the preprocessed actor dataset. Each clip featured a clearly visible face and was previously labeled with both the emotion type and its intensity level. Within Unity, a Skinned Mesh Renderer component was used to apply and manipulate blendshape weights on the avatar. This component enables smooth deformation of the 3D facial mesh, making it suitable for real-time animation and emotional transitions. Using Unity's interface, each of the 15 samples per emotion was manually matched to corresponding blendshape configurations by adjusting weight values for the avatar's facial features. These weights were recorded for every expression and grouped by intensity:

- Level 1 — Weak
- Level 2 — Moderate
- Level 3 — Strong

For each emotion-intensity pair, the corresponding blendshape values were averaged across all 15 samples. The resulting mean values were saved into a CSV file that mapped each emotion and intensity level to a fixed blendshape vector. This static mapping served as the core lookup table used to drive facial expression rendering in Unity during storytelling.

C. Emotion Receiving System in Unity

To manage emotion playback during runtime, a dedicated C# script named *EmotionReceiver.cs* was developed in Unity. This script stored the precomputed blendshape averages for all emotion-intensity combinations and acted as the main controller for updating the avatar's expressions. When triggered by other Unity components, such as the story playback script, the emotion receiver script applied the appropriate blendshape configuration to the Skinned Mesh Renderer. It also managed two additional features: automatic blinking every five seconds and smooth transitions between emotions using linear interpolation. These enhancements contributed to a more lifelike and continuous emotional animation experience. The entire process was driven internally within Unity through method calls between modular

C# scripts, allowing for flexible and reusable control over avatar behavior.

D. Audio Playback and Jaw Movement

Speech playback and jaw motion were handled by a dedicated Unity C# script named AvatarStoryPlayer.cs. This script managed the sequential playback of audio clips and controlled the avatar's jaw movement to simulate realistic speech. For each sentence in the story, the associated audio clip was loaded and played, while the corresponding emotion key was sent to the EmotionReceiver.cs script to trigger the appropriate facial expression. To animate the jaw in sync with speech, the script utilized Unity's GetOutputData() function to access raw audio samples in real time. The Root Mean Square (RMS) value of these samples was computed to estimate loudness, which was then mapped to the jaw blendshape weight. As the audio plays, the jaw moves dynamically in proportion to the volume, creating a more lifelike speaking behavior.

E. Final Integration

The avatar's expressive behavior results from a coordinated pipeline that combines audio, facial animation, and real-time emotion rendering. The integration is achieved through the following sequence:

- 1) A story script supplies both an AudioClip and a corresponding emotion key.
- 2) AvatarStoryPlayer.cs plays the audio and communicates the emotion key to EmotionReceiver.cs.
- 3) The corresponding facial expression is applied using blendshape values and interpolated smoothly.
- 4) The avatar blinks automatically at regular intervals to reduce visual rigidity.
- 5) The jaw motion is animated in real time based on the RMS loudness of the audio signal.

This integrated system enables the avatar to deliver emotionally expressive storytelling performances that align with both the cultural and linguistic expectations of Egyptian users. The result is a smooth and engaging experience designed to improve relatability and interaction, especially for children or individuals with communication challenges.

The system achieves an average end-to-end latency of 80–120 ms with stable playback at 55–60 FPS on mid-range hardware. Under higher computational load, the frame rate may drop to 40–50 FPS. Blendshape transitions are smoothed with a 0.5 s interpolation time to ensure natural facial animation.

V. SUBJECTIVE EVALUATION

To assess the effectiveness of emotional expression in the avatar, a structured user study involving 40 participants was conducted. The goal was to evaluate whether emotional facial expressions had an observable impact on participants' perceptions, comprehension, and engagement, while minimizing potential bias toward the study's objective.

A. Evaluation Design

The evaluation was conducted using six short video clips, each featuring the same avatar discussing different

perspectives on the topic of online gaming. Three of the videos included full emotional expressions, based on pre-labeled emotion and intensity annotations, while the other three featured a neutral avatar with no emotional cues. To reduce order bias, the videos were presented in randomized order for each participant. Half of the participants viewed the emotionally expressive videos first, and the other half viewed the neutral ones first. The study was designed to keep participants unaware that the research focus was on avatar expressiveness, ensuring more authentic and unbiased responses.

B. Participant Demographics

At the beginning of the questionnaire, participants were asked general demographic questions. These questions included:

- Age and gender
- Frequency of video game usage
- Personal views on online gaming

These initial questions aligned with the topic of the videos and helped maintain participant focus on the story content rather than the avatar's visual presentation.

C. Main Questionnaire

After viewing all six videos, participants were asked to respond to a series of statements using a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). These questions were carefully designed to indirectly assess the effectiveness of the avatar's emotional expressions, without revealing the purpose of the evaluation too early. The statements focused on: The evaluation focused on three key aspects: clarity, assessing whether participants found the story easy to follow; naturalness, evaluating whether the avatar's expressions appeared believable and contextually appropriate; and expressiveness, measuring how engaging or interesting the avatar made the viewing experience. In the final section of the questionnaire, participants were informed that the evaluation focused on emotional realism in avatars. They were then asked directly which version of the avatar they preferred (emotional vs. non-emotional), whether emotions enhanced their understanding, and how such avatars could support children's education. Feedback was also gathered on areas for improvement. All responses were collected anonymously to encourage honest opinions.

VI. RESULTS

A. Subjective Evaluation

A total of 40 participants took part in a structured evaluation aimed at understanding the impact of emotional expressions in the avatar. Each participant viewed six videos—three emotionally expressive and three neutral—featuring the avatar discussing the same topic with identical scripts. The order of video presentation was randomized to minimize order bias, and participants were initially unaware that the study involved testing emotional performance. The questionnaire was divided into three main parts: general demographics and topic context, a post-viewing rating section using a 5-point Likert scale, and a final direct comparison between the expressive and neutral avatars. Key findings are that the evaluation results showed a strong preference for the emotionally expressive avatar, with 95% of participants (38

out of 40) favoring it over the neutral version. Additionally, 82% of participants believed that an expressive avatar would have made educational storytelling more enjoyable during their childhood. The expressive avatar was consistently rated higher across three main dimensions: clarity, as it helped participants follow the story more easily; naturalness, as its emotions appeared believable and appropriate; and expressiveness, as it made the viewing experience more engaging and realistic. Qualitative feedback also highlighted that emotional expressions enhanced attention and made the avatar feel more “human” and relatable. This aligns with the goal of using emotional avatars to increase user immersion, especially in educational or therapeutic contexts. The results validate the importance of integrating emotional cues into avatar-based communication systems and demonstrate that emotional variability—especially when culturally grounded—can significantly enhance user engagement and perception of realism.

B. Visual Demonstration of Emotion and Intensity

To illustrate the expressive capabilities of the avatar, six core emotions—angry, happy, sad, fear, surprise, and neutral—were rendered at three different intensity levels: weak, moderate, and strong (where applicable). The images below demonstrate smooth blendshape transitions and visually distinguishable emotional intensities.

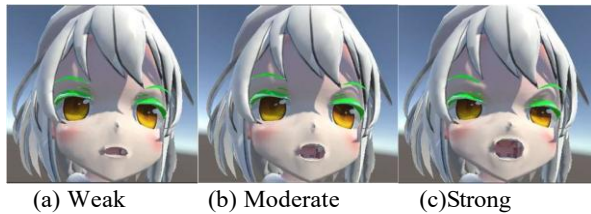


Fig. 1. Angry emotion at three intensity levels.

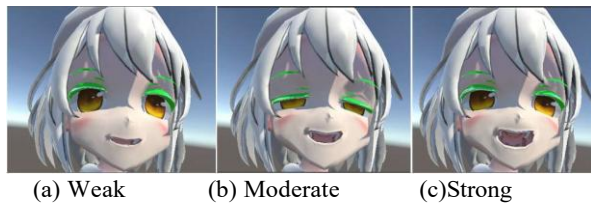


Fig. 2. Happy emotion at three intensity levels.

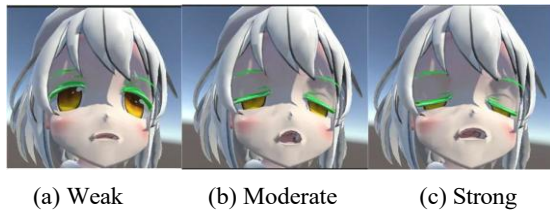


Fig. 3. Sad emotion at three intensity levels.

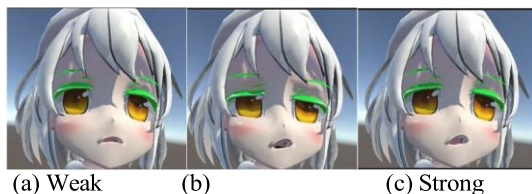


Fig. 3. Fear emotion at three intensity levels.

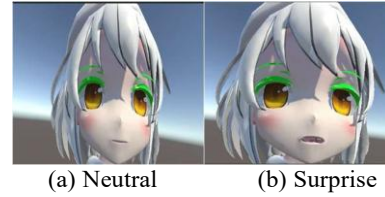


Fig. 4. Comparison between neutral and surprise

Additional realism was achieved through blinking and jaw movement synchronized with speech. A custom dataset of 207 emotion-labeled video clips from an Egyptian actor was used to guide avatar expression design and train a facial emotion recognition model for future integration. The final system was evaluated through a user study involving 40 participants, which demonstrated a strong preference (95%) for the emotionally expressive avatar. Participants reported increased realism and engagement, and 82% believed such a system would have improved their learning experience as children. While the avatar effectively expressed emotions, one key limitation was its lack of cultural visual identity, which affected relatability. Future work will focus on designing a more authentic Egyptian-looking avatar and incorporating dynamic emotion input—such as detecting the user’s emotion from an image and generating a matching story. These improvements aim to enhance interactivity, cultural alignment, and user immersion, advancing the avatar toward a fully responsive, emotionally aware storytelling agent.

VII. CONCLUSION AND FUTURE WORK

This work presented the design and development of an emotionally expressive avatar capable of narrating Egyptian Arabic stories with synchronized facial expressions and speech. The system supported six core emotions—happy, sad, angry, fear, surprise, and neutral—each expressed at three intensity levels. Emotional expressions were created by averaging blendshape configurations manually matched from real actor samples.

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